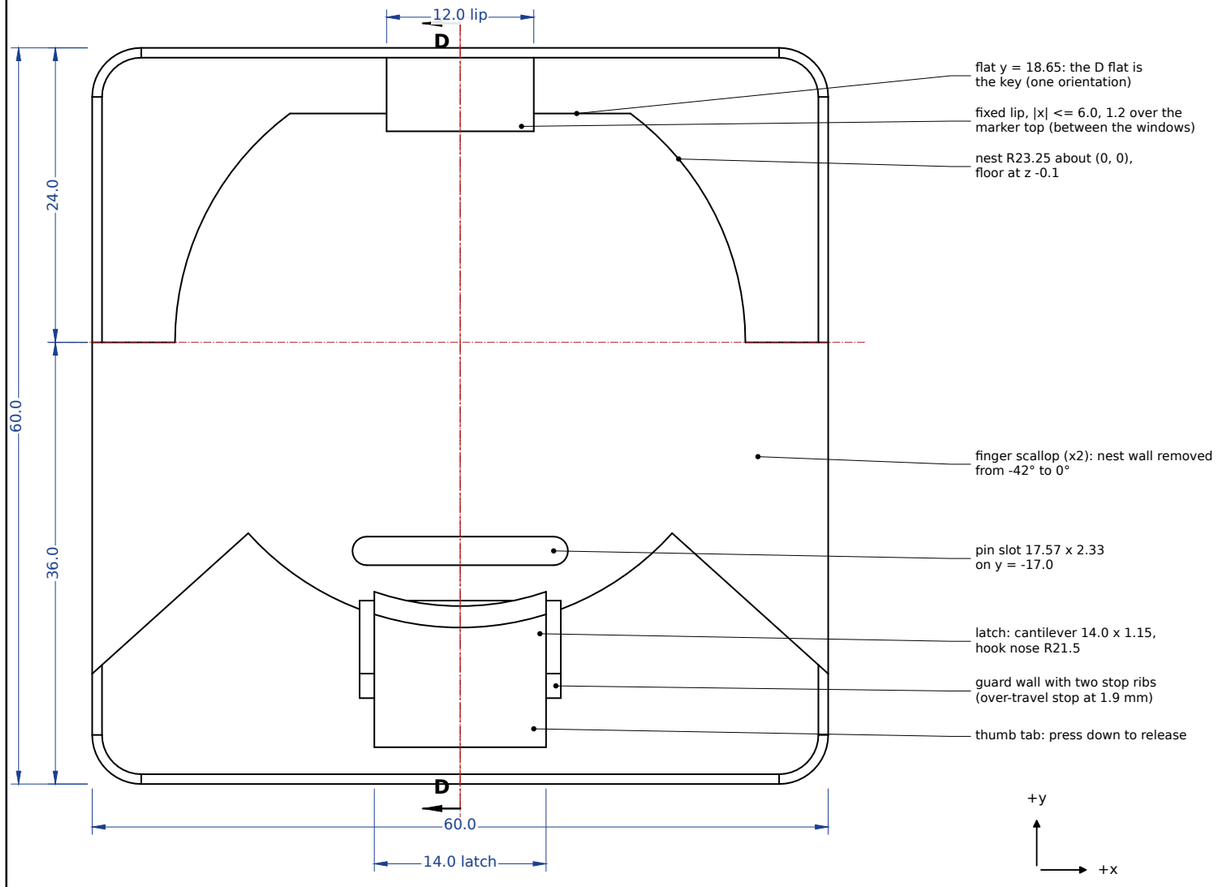
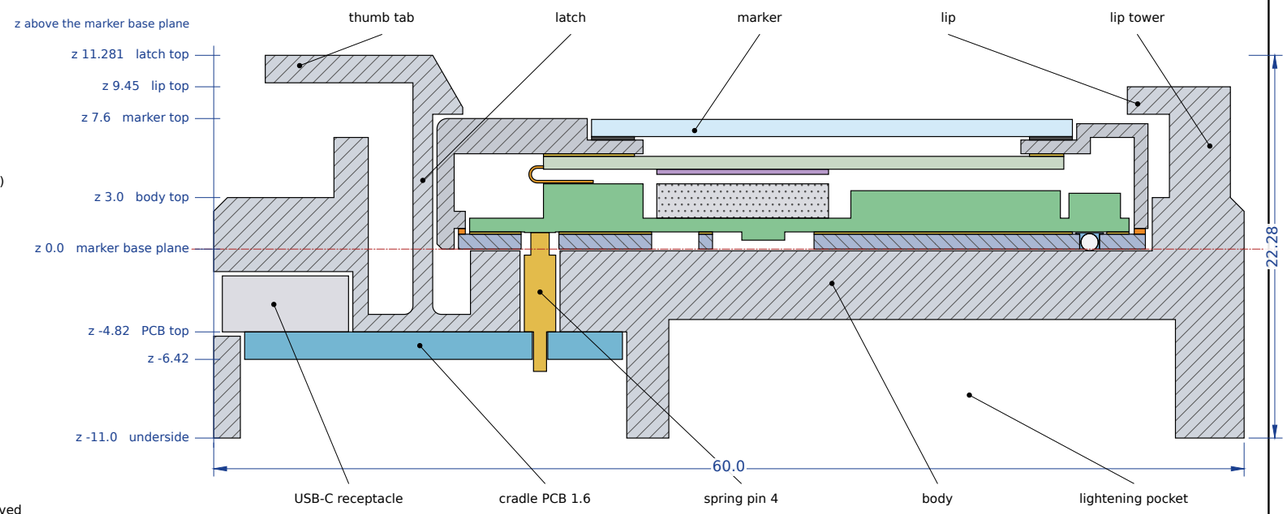
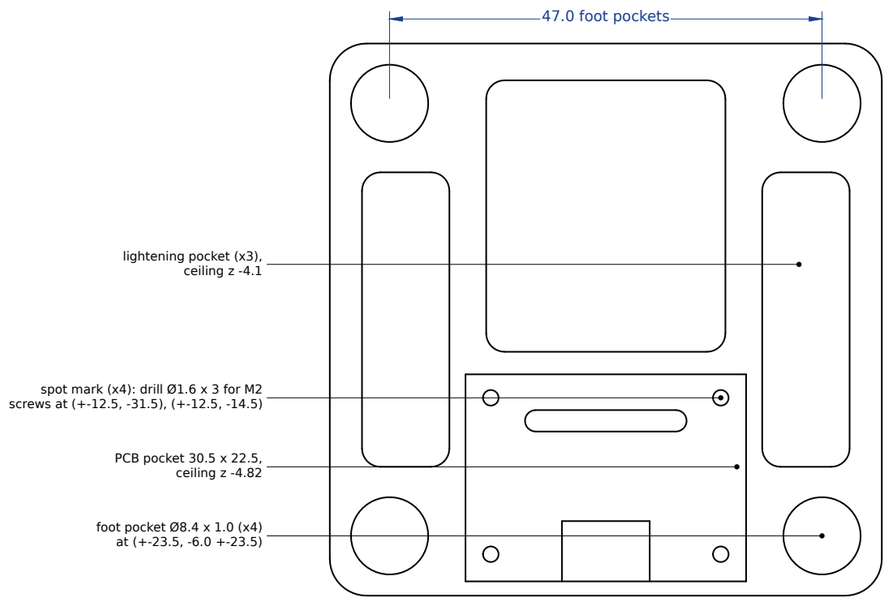
TOP VIEW 2:1



SECTION D-D (x = 0), marker seated, centre pin on its pad 2.8:1



VIEW FROM BELOW 1.5:1 (+x to the LEFT)



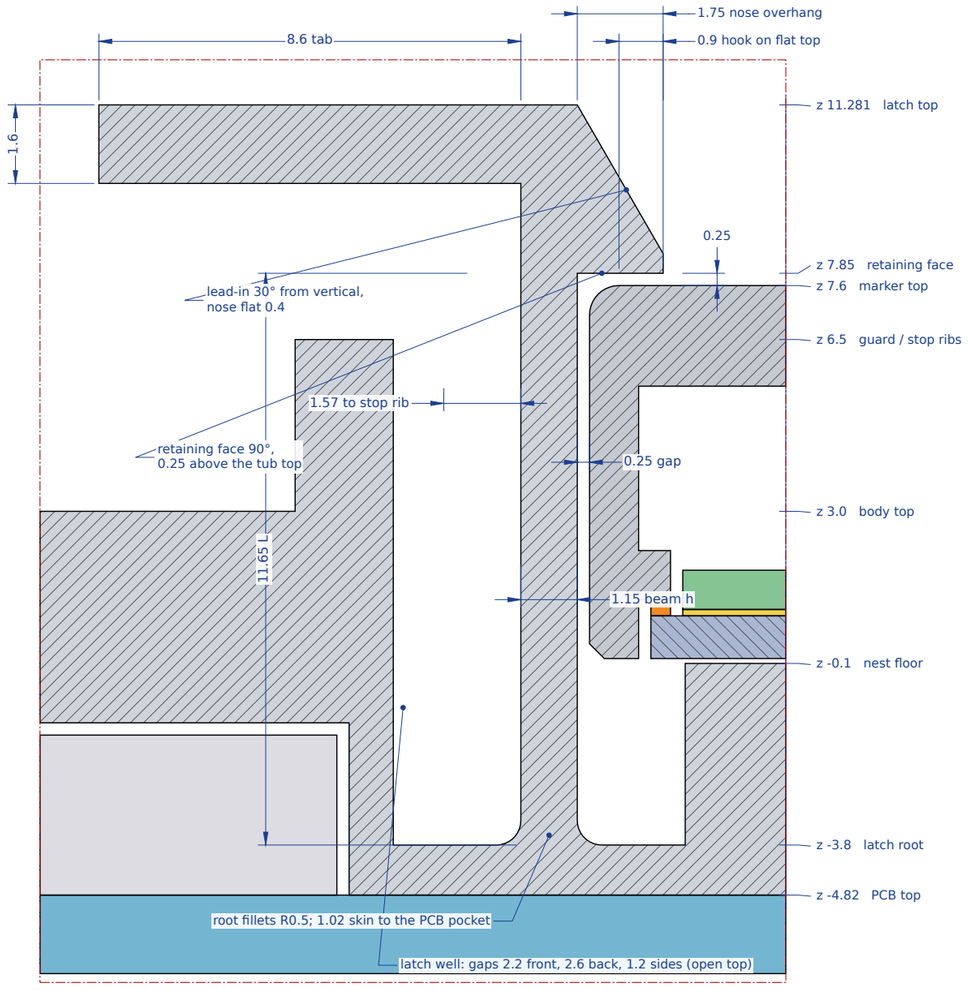
NOTES

1. Rev C: charging and data dock only. Printed body 60.0 x 60.0 x 14.0 mm (z -11.0 to 3.0); lip and latch stand above it, overall height 22.3 mm (D20). Frame: the marker frame with the marker seated (origin at the marker centre on its base plane).
2. The nest is the marker outline + 0.25 mm; the D flat is the key, so the marker seats one way round only. It rests on its three ball feet on the nest floor (z = -0.1). No reference plate, no 180° position. Zeroing does not use the cradle.
3. Hold-down: fixed lip over the top front edge (|x| <= 6.0, clear of windows, cover glass and vent) + sprung latch at the rear with a thumb tab. The marker goes in nose first under the lip and is pressed down at the rear until the latch clicks. Details and numbers: sheet 2.
4. Cradle PCB top at z = -4.82: the spring pins stand at their working height (5.82) on the pad plane z = 1.0. Seven pin positions, consumer build fits four (VBUS, D-, D+, GND at x = -7.62 ... 0.0), factory fixture seven. Pin and receptacle envelopes are PROVISIONAL.
5. Print with the latch beam lying in the build plane if the bureau agrees; no wall below 0.8 mm, latch root 1.15 mm, no trapped powder (checks 5, 6).

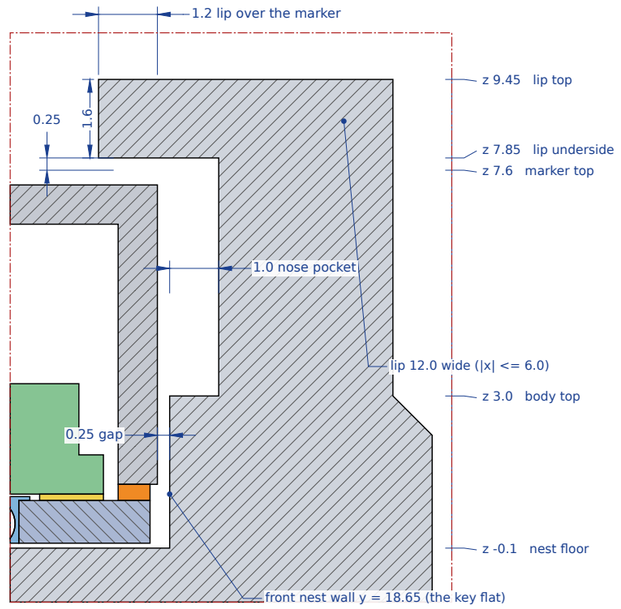
Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE: **GL1-020 Cradle (charging / data dock with hold-down)**		
PROJECT / PART: GreenLine GL-1 electronic ball marker - cradle body		
MATERIAL / FINISH: PA12 (HP 3D HR PA 12), any colour	PROCESS: HP Multi Jet Fusion; drill 4 x Ø1.6 after printing	
GENERAL TOLERANCE: printed: +-0.3 or +-0.3 % (bureau MJF), as printed	SCALE (A3) / UNITS: 2:1 / 2.8:1 / 1.5:1 mm	
STATUS: **PROVISIONAL - not released**	DWG REV / PARAMS: C-EVT1 / C-EVT1	SHEET / DATE: 1 of 2 2026-11-11

DETAIL J - rear latch, section x = 0 8:1

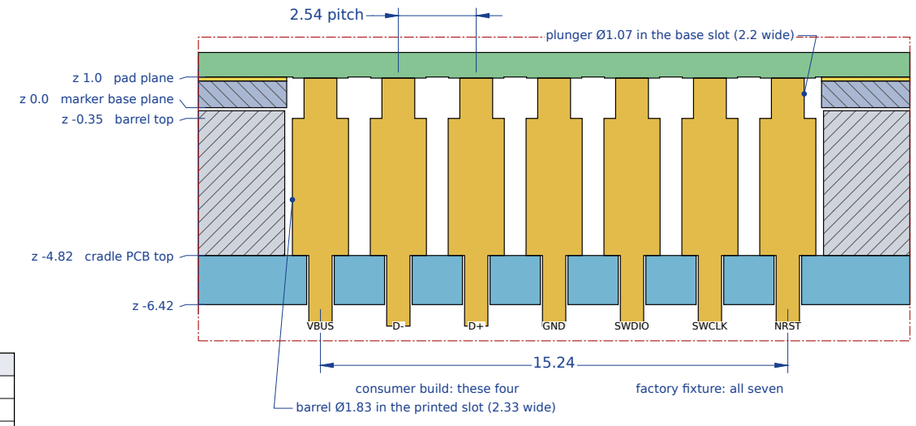


DETAIL K - front lip, section x = 0 8:1

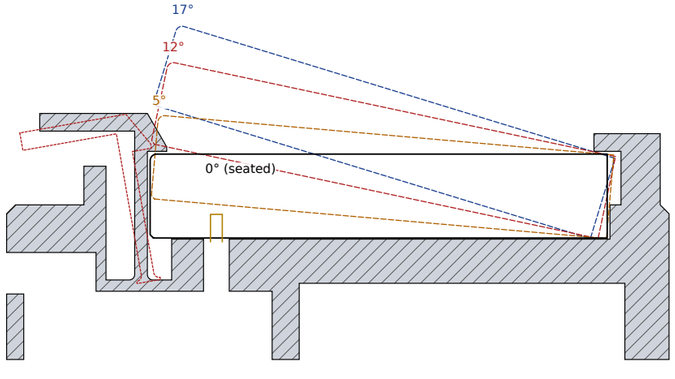


Cradle PCB (2 layers)	Value
Outline	30.0 x 22.0 x 1.6, x +-15.0, y -34.2 to -12.2
Pin positions (7)	y = -17.0, x = -7.62 ... 7.62, pitch 2.54
Screws (4 x M2 thread-forming)	(+-12.5, -31.5) and (+-12.5, -14.5)
USB-C receptacle	GCT USB4105, top face, mouth at y -35.5
Top face	z = -4.82 (pad plane 1.0 - working height 5.82)

SECTION E-E (y = -17.0) pin row, marker seated 5:1



INSERTION PATH 1.8:1 section x = 0, marker outline at four tilts



Hold-down (PA12 MJF, E = 1700 MPa); check 7	Value
Latch beam b x h x L (uniform cantilever)	14.0 x 1.15 x 11.65 mm, root 1.15 >= 1.0
Pin force: 7 pins / 4 pins; marker weight	4.13 N / 2.36 N; 0.11 N
Retention, no friction: F = 2 E I u / (e L^2)	21.3 N at the hook = 5.9 x the 7-pin force (>= 3 x)
marker fully forward in the nest (u = 0.65)	14.4 N = 4.0 x; self-locking above mu 0.24
Hook travel: insertion / release / stop	1.36 / 1.50 / 1.90 mm
Root strain 3 h d / (2 L^2) at those travels	1.72 / 1.91 / 2.41 % (limit 2.5 %)
Insertion force at the thumb (mu 0.2 / 0.3 / 0.4)	8.5 / 10.2 / 12.2 N
Release: thumb on the tab (nominal / worst)	8.7 / 10.1 N
Marker float against lip and hook with pins fitted	up to 0.25 mm (pins then at 6.07 mm)

NOTES

1. Tilt the marker nose-down, slide the top front edge under the lip, lower the rear: its bottom edge runs down the 30° lead-in, pushes the latch back by up to 1.36 mm (red, dotted) and the hook snaps over the top.
2. The pads meet the free plungers (tip z 2.14, orange) at about 1.9° of tilt; seated, the pins are compressed 1.14 of 2.29 mm.
3. Release: press the thumb tab down; the pins lift the rear about 1.3 mm; lift the marker at the two finger scallops.
4. Check 7 poses the marker solids at 19 tilts from 17° to 0°: no interference with lip, nest walls, floor or pins.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE

GL1-020 Cradle (charging / data dock with hold-down)

PROJECT / PART

GreenLine GL-1 electronic ball marker - hold-down details, pin row, insertion

MATERIAL / FINISH	PROCESS	
PA12 (HP 3D HR PA 12), any colour	see sheet 1	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	8:1 / 5:1 / 1.8:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	C-EVT1 / C-EVT1	2 of 2 2026-11-11